Prepared by group 9

Group Project

Hoax News classification using IndoBERT

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Background & Problem Statement

The Context:

Indonesia's rapid surge in internet accessibility has unfortunately outpaced digital literacy, where hoaxes spread significantly faster than valid news

The Gap

Standard multilingual models (like mBERT) often lack the deep cultural and understanding required to detect sophisticated Indonesian hoaxes.

Solution

Using IndoBERT, we can accurately extract contextual meanings from Indonesian text and improve our NLP model's performance.

IndoBERT

Why IndoBERT

Utilizes BERT (Bidirectional Encoder Representations from Transformers), which uses Self-Attention to understand a word's meaning based on its surrounding context (left and right).

Mechanism

We apply Transfer Learning, fine-tuning this pre-trained "brain" specifically for binary classification (Hoax vs. Non-Hoax).

Dataset & Preprocessing Pipeline

https://www.kaggle.com/datasets/muhammadghazimuharam/indonesiafalsenews

Noise Removal:

Utilized Regex to strip URLs (http://...) and punctuation, followed by Case Folding to lowercase all text.

Stopword Removal:

Applied the Sastrawi library (StopWordRemoverFactory) to remove high-frequency, low-meaning particles (e.g., 'yang', 'dan') to reduce noise.

Tokenization:

Converted text to tensors using BertTokenizer with a max sequence length of 64 tokens to optimize GPU memory usage.

Implementation & Hyperparameters

Framework:

- PyTorch
- Hugging Face Transformers.

Training Dynamics

- Batch Size: 16 samples per step.
- Epochs: 2 Epoch (13.7 minutes runtime).

Optimizer

- AdamW (Adam with Weight Decay) to handle sparse gradients.
- Learning Rate: 3e-5 (Optimal range for BERT fine-tuning).

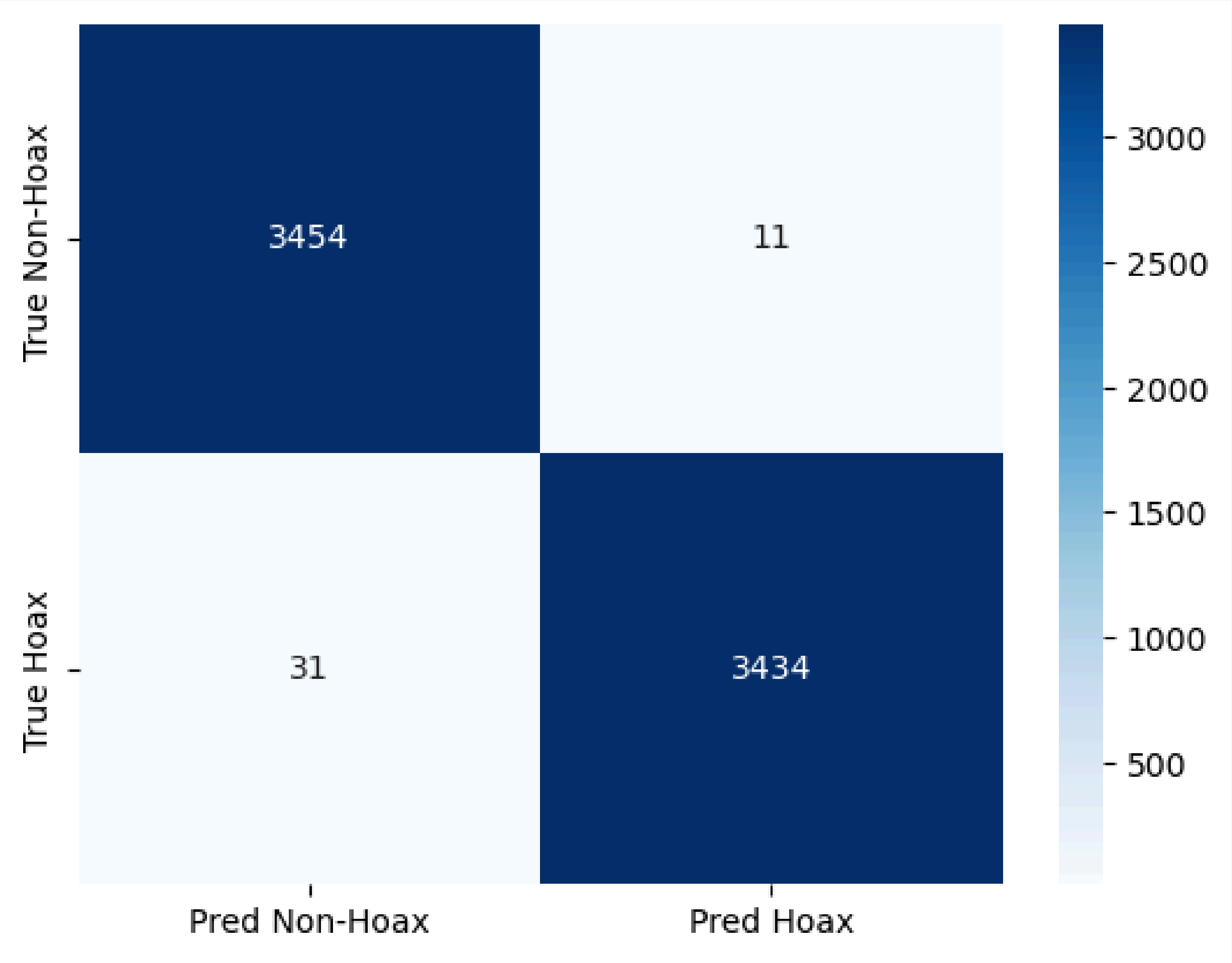
Optimization

- Implemented Mixed Precision Training (GradScaler) to accelerate backpropagation and reduce VRAM consumption without losing model accuracy.

Quantitative Results

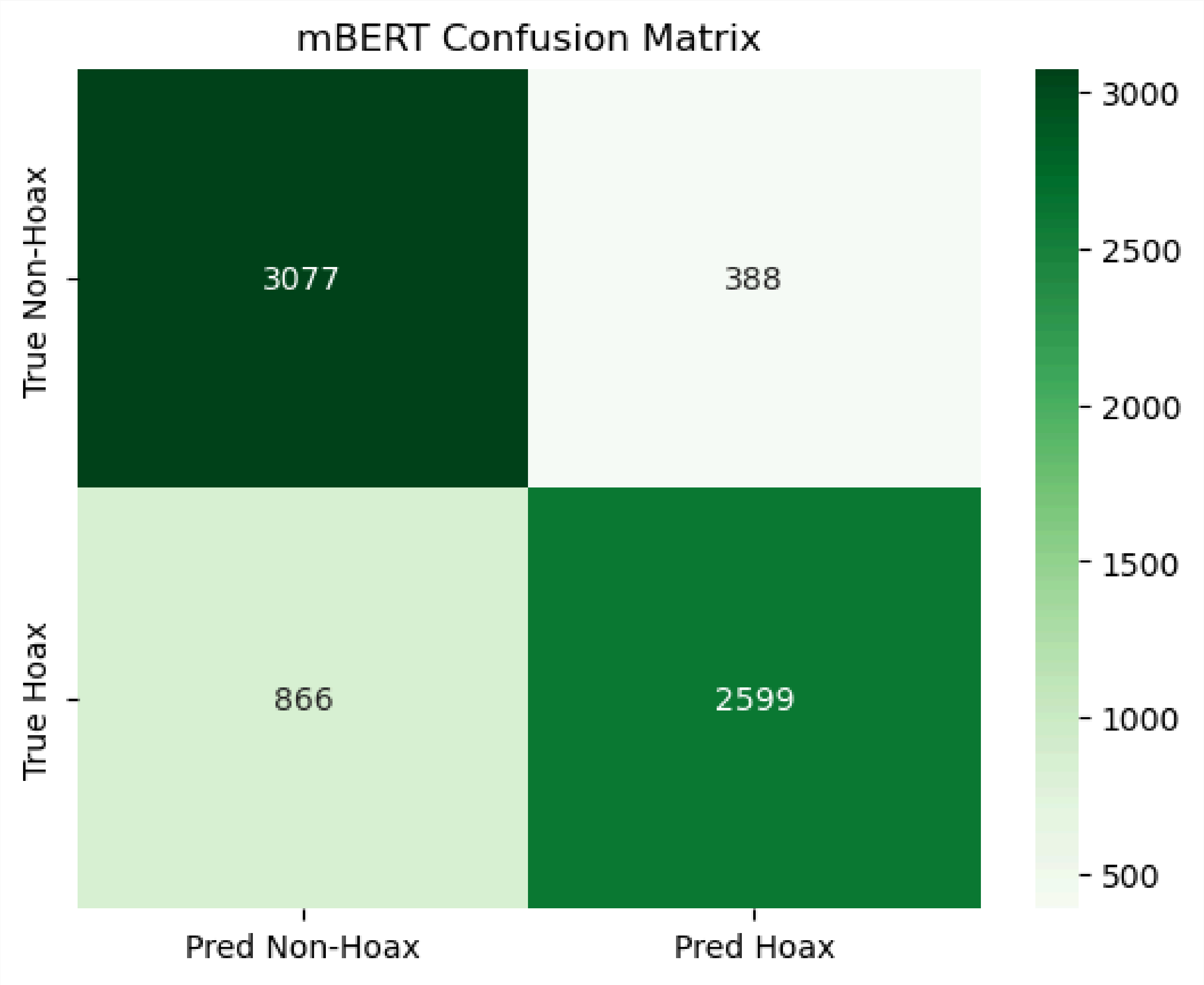
IndoBERT (Indonesia's BERT)

- Overall Accuracy: 99%
- Precision: 1.00
- Recall: 0.99



mBERT (Multilingual BERT)

- Overall Accuracy: 99%
- Precision: 1.00
- Recall: 0.99



IndoBERT vs. mBERT Comparison

	IndoBERT	mBERT
Training Time	~ 5 minutes	~ 45 minutes
Language Focus	Specialized (Indonesia)	Generalized (104 Languages)
False Negative	31	866
Vocabulary Size	~ 30k - 50k	~ 119k

While both models have similar architectures, IndoBERT is far more reliable. It produced significantly fewer False Negatives, meaning it almost never lets a Hoax slip through undetected.

Conclusion

IndoBERT is the superior choice for local news classification. It is 9x faster to train and achieved 99% accuracy compared to mBERT's 82%

Specialized monolingual models outperform generalized multilingual models because they understand local nuance/slang (context) better than "Jack-of-all-trades" models like mBERT.

Future improvements could include testing IndoRoBERTa and expanding the dataset to include regional dialects (Javanese/Sundanese).

Reference

- Wilie, B., et al. (2020). "IndoNLU: Benchmark and Resources for Evaluating Indonesian Natural Language Understanding." Proceedings of the 1st Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (AACL 2020)
- Conneau, A., et al. (2020). "Unsupervised Cross-lingual Representation Learning at Scale." Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL 2020)
- Purwarianti, A., & Crisdayanti, E. (2021). "Benchmarking IndoBERT and Transformer Models for Sentiment Classification on Indonesian E-Government Service Reviews.
- Koto, F., et al. (2020). "IndoLEM and IndoBERT: A Benchmark Dataset and Pre-trained Language Model for Indonesian NLP.

Thank you